

Question Paper

Exam Date & Time: 28-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
EIGHTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Advanced Data Structure [IT_5013H]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Derive the worst-case time complexity of the Rec function (5)
- A)
- ```
int Rec(int n) {
 int count = 0;
 for (int i = 1; i <= n; i++) {
 if (i % 2 == 0)
 count += Rec(i / 2);
 }
 return count + 1;
}
```

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Apply backward substitution to enable the time complexity for the recurrence relation  $T(n) = 2T(n/2) + n \log n$ . (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) Show the step-by-step process of the Insertion Sort algorithm to sort the following numbers (5)  
42, 21, 55, 2, 12, 32
- A) State the worst-case time complexity of the algorithm and explain why it is the worst case.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Given the following sequence of integers: 50, 30, 70, 20, 40, 60, 80 (5)
1. Construct a Binary Search Tree (BST) by inserting the elements in the given order.
  2. Draw the final BST structure.
  3. Explain the in-order, pre-order, and post-order traversals of the constructed BST.
  4. Analyze the time complexity of search, insertion, and deletion operations in a BST in both average and worst-case scenarios.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3) "wearetryingtolearndesignandanalysisofalgorithm" for the given phrase, construct the Huffman coding and generate the code for "Tester". Compare the length of code between fixed and variable length code for the given word "Tester". (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Given a set of positive integers {3,2,5,7,6,8,10,4} and a target value "20", determine whether some subset has a sum equal to the target. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4) Consider the problem with  $n = 4$ ,  $V = \{10, 10, 12, 18\}$ ,  $w = \{2, 4, 6, 9\}$  and  $W = 15$ . What is the highest profit that might be made utilising the Branch and Bound strategy using the above-mentioned 0/1 Knapsack? (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Use a Genetic Algorithm to solve a simple optimization problem. Apply selection, crossover, mutation, and fitness evaluation to evolve the population over multiple generations. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Apply the concepts of  $P = NP$  and  $P \neq NP$  to real-world scenarios in cryptography, optimization, AI, and software verification. Use specific examples to demonstrate how each outcome would affect problem-solving in these fields. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Assume a polynomial-time algorithm exists for the Hamiltonian Cycle problem. Apply polynomial-time reductions to transform other NP problems into Hamiltonian Cycle and demonstrate how they could be solved using this algorithm. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

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| Event Name                        | Subject                          |
|-----------------------------------|----------------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025-2026 | IT_5013H:Advanced Data Structure |

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 15    |
| 2     | 2              | CO 2        | 10    |
| 3     | 3              | CO 3        | 10    |
| 4     | 4              | CO 4        | 15    |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

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|-----------------------------------|----------------------------------|
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| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 5               | 25(50%)   |
| 4     | Analysis    | 5               | 25(50%)   |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 10              | 50 (100%) |